

Migrate SQL Server On-prem DB to Azure || Multiple Ways

The main strategies to move a 1 TB on-prem SQL Server to Azure with minimal downtime.

We'll cover both Azure SQL Managed Instance, Azure SQL Database, and SQL Server on Azure VM (IaaS).

1. Migration to Azure SQL Server on VM (IaaS)

This is similar to AWS EC2 – you manage SQL Server yourself.

Option A – Backup & Restore (Native)

Steps:

1. Take a **full backup** on-prem.
2. Copy **.bak** to Azure Blob Storage (via AzCopy or Azure Storage Explorer).
3. Restore backup on Azure VM SQL Server.
4. For reduced downtime:
 - After full restore, take **differential** and **transaction log** backups.
 - Restore them with **NORECOVERY**.
 - At cutover, take **tail-log backup**, restore with **RECOVERY**.

Downtime: Minutes–hours (depending on final log restore + app switch).

Good for: Medium bandwidth, simple cutover.

Option B – Log Shipping

Steps:

1. Restore full backup to Azure VM with **NORECOVERY**.
2. Ship logs regularly from on-prem to Azure VM.
3. At cutover, stop writes, send final log backup, restore, and bring DB online.

Downtime: Minutes.

Good for: Large DBs (1 TB+) needing minimal downtime.

Option C – Always On Availability Group (Hybrid)

Steps:

1. Set up VPN or Azure ExpressRoute.
2. Create **Always On AG** between on-prem SQL Server and Azure VM SQL Server.
3. Let it sync.
4. Perform failover to Azure.

Downtime: Seconds to minutes.

Good for: Mission-critical migrations, but requires Enterprise Edition.

2. Migration to Azure SQL Managed Instance (PaaS)

Supports near-full SQL Server compatibility and allows native backups.

Option A – Native Backup/Restore to Azure Blob

Steps:

1. Take full backup on-prem to **.bak**.
2. Upload to Azure Blob.
3. Use **T-SQL RESTORE** on Managed Instance (supports **WITH MOVE**).
4. Use **log backups** for near-zero downtime cutover.

Downtime: Minutes to hours depending on final restore.

Good for: Lift-and-shift with minimal changes.

Option B – Azure Database Migration Service (DMS)

Steps:

1. Provision **Azure DMS**.
2. Configure **Online Migration** (full + incremental sync).
3. DMS copies schema/data and keeps syncing changes until cutover.
4. Stop writes on-prem, let final sync happen, switch apps.

Downtime: Minutes.

Good for: Low downtime migrations without complex manual steps.

Option C – Transactional Replication

Steps:

1. On-prem SQL Server = Publisher, Azure SQL Managed Instance = Subscriber.
2. Snapshot initial data to MI.
3. Let replication keep it synced.
4. Cut over after final sync.

Downtime: Seconds–minutes.

Good for: Continuous sync until planned cutover.

3. Migration to Azure SQL Database (PaaS, Single/Elastic Pool)

Since Azure SQL Database is not fully SQL Server, methods differ.

Option A – DMS Online Migration

- Works like with Managed Instance, but requires schema compatibility.
- Supports continuous sync until cutover.

Option B – Transactional Replication

- Azure SQL Database supports being a subscriber (with limitations).
- Good for near-zero downtime migrations.

Option C – BACPAC Export/Import (High Downtime)

- Export DB to **.bacpac** file, upload to Azure Storage, import into Azure SQL Database.
- **Not for minimal downtime** (requires downtime during export/import).

4. Low Bandwidth Large DB Special Case

For 1 TB+ DBs over slow internet:

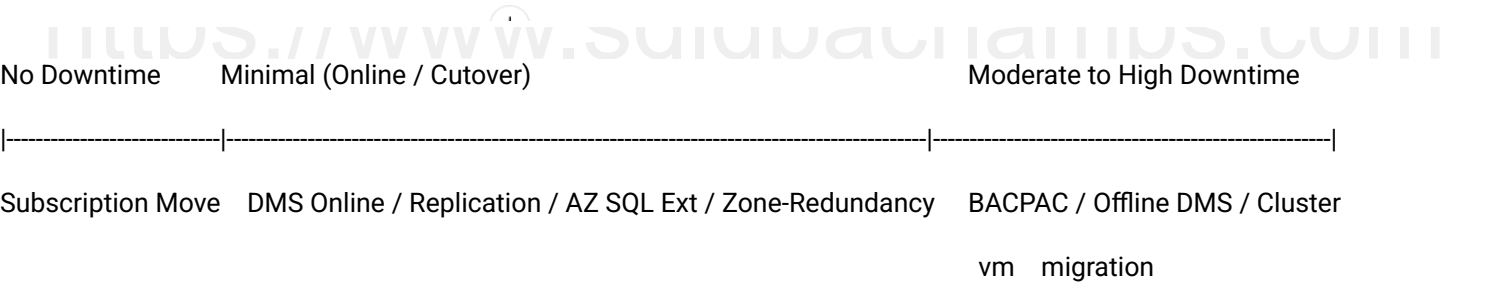
- Use **Azure Data Box** (similar to AWS Snowball):
 1. Backup DB to Data Box device.
 2. Ship to Azure.
 3. Azure uploads to Blob Storage.
 4. Restore in Azure VM or Managed Instance.
 5. Sync final diffs/logs over network before cutover.

Summary Table

Method	Target	Downtime	Complexity	Notes
Backup + Diff/Log Restore	VM/MI	Min–Hours	Low	Easy, manual cutover
Log Shipping	VM	Minutes	Medium	Reliable for large DBs
Always On AG	VM	Seconds–Minutes	High	Needs VPN/Enterprise
Native Backup/Restore to Blob	MI	Min–Hours	Low	PaaS-compatible
DMS Online Migration	MI/SQL DB	Minutes	Medium	Schema pre-check needed
Transactional Replication	MI/SQL DB	Seconds–Minutes	Medium	Limited features in SQL DB
Data Box + Diff/Log	VM/MI/SQL DB	Minutes–Hours	Medium	For slow bandwidth

Azure Database Migration Paths & Downtime Overview

Migration Path	Description / Tools	Downtime Window
1. Azure Database Migration Service (DMS)	Supports <i>offline</i> (one-time) and <i>online</i> (continuous sync) migrations; works for SQL Server, MySQL, PostgreSQL, MongoDB → Azure SQL Managed Instance, Azure SQL VM, etc. Microsoft Azure Microsoft Learn	Offline: downtime starts at migration start Online: minimal downtime – only during final cutover Microsoft Learn +2
2. Transactional Replication (SQL Server → Azure SQL DB)	Uses SQL Server transactional replication to continuously sync data; switch over when ready prosperasoft.com Red9	Minimal—only during cutover
3. BACPAC Export/Import	Export schema & data to BACPAC, import into Azure SQL. Often used for small/offline migrations Red9 Microsoft for Developers	High (downtime during export/import)
4. Azure SQL Managed Instance Extension (via Azure Data Studio)	Supports online migrations with minimal downtime, especially from SQL Server sources Microsoft Learn Reddit	Minimal—final switchover only
5. Subscription/Resource Move (Azure SQL server)	Moving Azure SQL across subscriptions or resource groups has no downtime —applications remain online (possible short lock) Microsoft Learn	None—negligible lock (typically < 4 hours)
6. Zone-Redundancy Migration	Switching Azure SQL tiers (e.g., Premium to zone-redundant) is an online operation , with only a brief disconnect at the end Microsoft Learn	Minimal—brief transient disconnect during final sync
7. VMware-Based Migrations (Always-On / Standalone)	Used for migrating SQL Server via VMware tools: Always-On AG minimally impacts uptime; standalone may have low downtime; cluster migrations involve high downtime GitHub	Varies from Low to High depending on topology



Recommendations by Scenario

- **Zero or near-zero downtime required:** Use **DMS online migration**, **transactional replication**, or the **Azure SQL extension via Data Studio**.
- **No downtime:** Move within Azure subscriptions or enable zone redundancy—both are effectively seamless.
- **Small workload or low tolerance for complexity:** BACPAC import may be acceptable—but expect downtime and test carefully.
- **Large / enterprise environment with Always-On setup:** VMware-based migration with AG minimizes service disruption.
- **Zone-resilience upgrade:** A fast, online process with minimal disconnect.

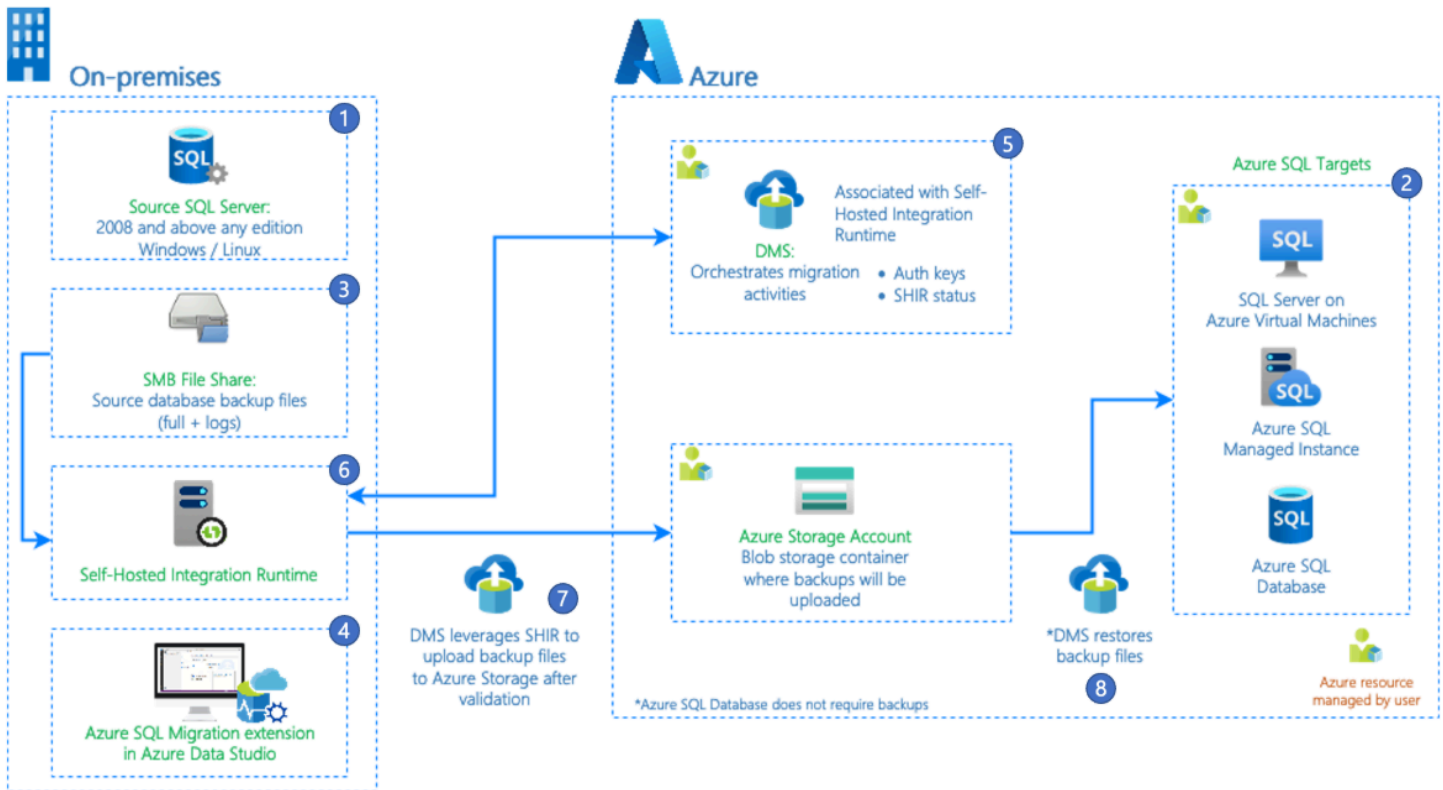


Diagram Overview (above)

The image shows end-to-end migration flows using tools like the Azure SQL Migration extension for Azure Data Studio and Azure Database Migration Service (DMS). It distinguishes migration scenarios—such as migrating from SQL Server to Azure SQL Managed Instance or Azure SQL Database—and indicates whether they support **online** (minimal downtime) or **offline** (full downtime) modes.

How to Leverage This Diagram

1. **Start with assessment** using Azure SQL Migration extension for Azure Data Studio:
 - It helps you discover migration readiness and recommends appropriate Azure target SKUs. [Microsoft Learn](#)
2. **Choose target & mode:**
 - If low-downtime is critical, opt for **online migration** to Managed Instance or SQL Server on VM.
 - If downtime is tolerable, **offline migration** offers simplicity.

3. Plan your strategy:

- Map out downtime windows and communicate cutover times.
- Test offline migration and iterate.
- Ensure infrastructure (like network, DMS configuration, backups) is in place for smooth cutover.

Migration Modes & Downtime Profiling

Migration Path	Mode	Downtime Characteristics
SQL Server → Azure SQL Managed Instance	Online	Downtime only during final cutover after continuous data sync—minimal downtime. Microsoft Learn +2 TECHCOMMUNITY.MICROSOFT.COM
	Offline	Downtime begins at migration start until completion. Includes full backup, data transfer, and restore. Microsoft Learn +1 BDRSuite TECHCOMMUNITY.MICROSOFT.COM
SQL Server → SQL Server on Azure VM	Online	Similar to Managed Instance: minimal downtime during final cutover. Microsoft Learn +2
	Offline	Downtime through the full duration of migration—backup, transfer, and restore. Microsoft Learn +1
SQL Server → Azure SQL Database	Offline	Only offline mode is supported; downtime starts at migration initiation. Microsoft Learn +2

Insights & Best Practices

- **Online migrations:** Designed to reduce business disruption by letting your source database remain live until the final cutover. Ideal for production environments. Requires a Premium-tier DMS instance.
- **Offline migrations:** Simpler to implement but require full downtime from start to finish. Best for development, lower-traffic environments, or when downtime is acceptable.
- **Recommendation:** Always conduct a test run with the offline migration to assess downtime. If it's unacceptable, switch to the online option.
- **Tool integration:** Leverage the Azure SQL Migration extension for assessment, right-sized SKU recommendation, and picking between migration modes in one guided experience.